Individual Semester Project : Milestone 1

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Project ID	CMPG325-2026-112
Client ID	CLI-112
Assigned Organisation	Dikgosi History & Archives Research Centre (Mafikeng)
Industry	Research
Assigned Addressing Block	10.42.0.0/16

1. Client Requirements

1.1 Client Background

The Dikgosi History & Archives Research Centre is a research organisation that catalogues, digitises, and provides access to historical records. Based on the assigned project brief, the network must support research and administrative staff and a public-facing reading room, while hosting an internal web service used for cataloguing and archive access.

1.2 Stakeholders Identified

Group	Description	Network Need
Administration	Finance, HR, front-office staff	General LAN + internet access
Archives & Research Staff	Historians, archivists, digitisation staff	LAN access + internal web portal (VLAN 40)
Reading Room (Public)	Walk-in researchers/patrons on public terminals	Isolated, internet-only; no access to staff/server VLANs
Off-site Administrator (CR9)	Network administrator working remotely	Secure (SSH) remote management of network devices only

Assumption: the brief does not specify exact headcounts. Two representative hosts per department are modelled in Packet Tracer; each VLAN's /24 subnet leaves headroom to scale to the real staff complement without re-addressing.

1.3 Functional Requirements

1. Full IP connectivity between all authorised departments, using the assigned block 10.42.0.0/16.
2. An internal HTTP/web server (assigned technical challenge) hosting an internal portal, reachable by Administration and Archives/Research staff only, not exposed to the public internet.
3. Logical segmentation (VLANs) so public Reading Room traffic cannot reach staff or server subnets.
4. Secure remote management access for one off-site administrator (CR9): SSH only, restricted by source address, Telnet disabled on all devices.
5. DHCP for end-user subnets (Admin, Archives, Reading Room); static addressing for infrastructure and the server.
6. A working, testable Cisco Packet Tracer implementation demonstrating the above.

1.4 Non-Functional Requirements

- Availability: no network downtime is tolerated during month-end processing (Design Constraint, §8 of brief), addressed through link redundancy on the internal distribution-access path (Section 2).
- Security: management access restricted and encrypted (SSH, ACLs); public and staff traffic kept in separate broadcast domains.
- Scalability: the /16 addressing block leaves substantial room for future departments or services.
- Maintainability: clear VLAN-to-department mapping and a consistent, documented addressing scheme.

1.5 Assumptions & Out of Scope

- Full WAN/ISP-side redundancy (dual ISP links, HSRP) is not implemented; the downtime constraint is addressed at the internal LAN layer, where cabling/port failures are the most common cause of local outages. Flagged as a recommendation for future work.
- The internal web server is not NAT-exposed to the internet; only the off-site administrator's management traffic (CR9) traverses the WAN link.
- Exact departmental headcounts, floor plan, and physical cabling lengths are not specified in the brief and have been assumed for this simulation.

2. Physical Topology

The physical topology below shows the devices, models, interfaces, and cabling used to implement the network. A collapsed-core design was chosen: an edge router handles only WAN/remote-access traffic, a multilayer core switch performs all inter-VLAN routing, and a separate access switch isolates the public Reading Room from staff and server traffic.

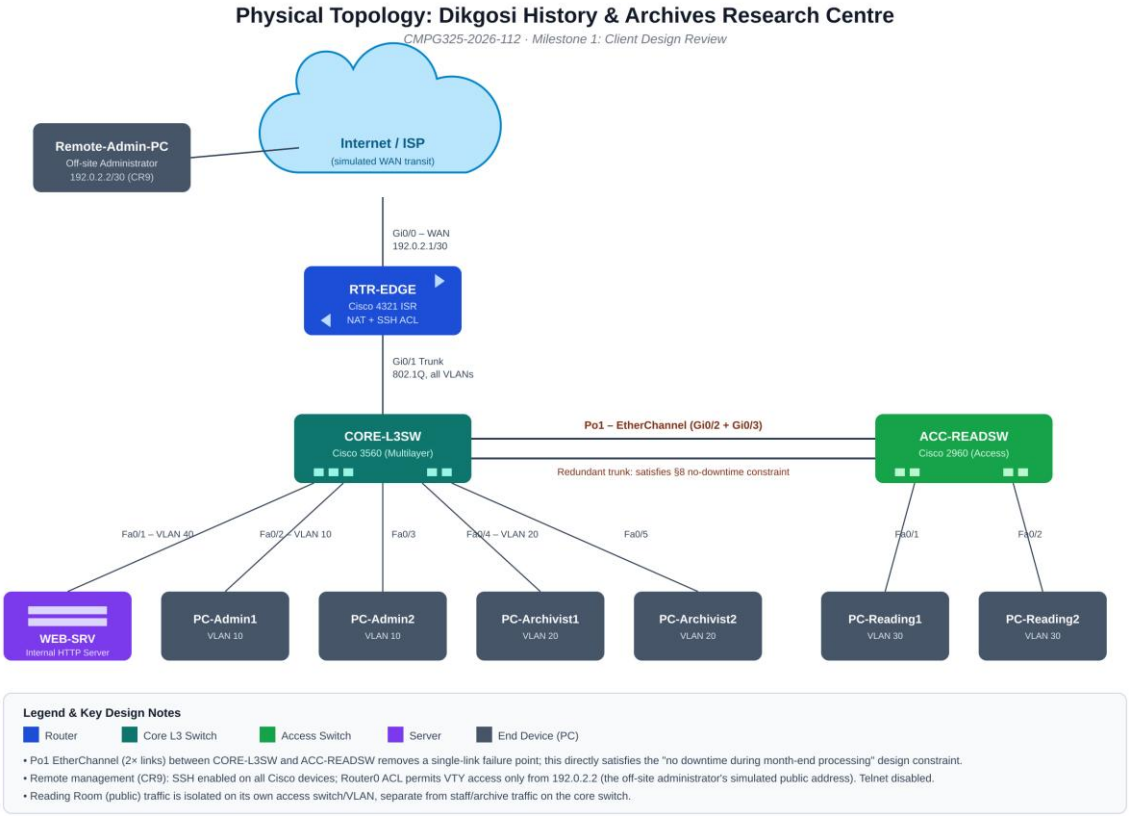


Figure 1. Physical topology: devices, interfaces, and cabling

2.1 Key Design Decisions

- RTR-EDGE (Cisco 4321 ISR) is the only device facing the WAN: it performs NAT/PAT for outbound internet access and terminates the secure remote-management path required by CR9.
- CORE-L3SW (Cisco 3560 multilayer switch) performs inter-VLAN routing via SVIs rather than router-on-a-stick, keeping local VLAN-to-VLAN traffic off the WAN-facing router entirely.
- ACC-READSW (Cisco 2960) is a dedicated access switch for the public Reading Room (VLAN 30), so public traffic never shares an access device with staff or archive traffic.
- Po1 EtherChannel (two bundled physical links between CORE-L3SW and ACC-READSW) removes a single-link failure point. This directly satisfies the brief's "no downtime during month-end processing" design constraint (§8) without the added complexity of a second router.
- Remote management (CR9): SSH is enabled on all Cisco devices; an ACL on RTR-EDGE permits VTY access only from 192.0.2.2, the off-site administrator's simulated public address. Telnet is disabled throughout.

3. Logical Topology

The logical topology shows the VLAN structure, IP subnets, and gateway addressing that sit on top of the physical design, independent of cabling and port numbers.

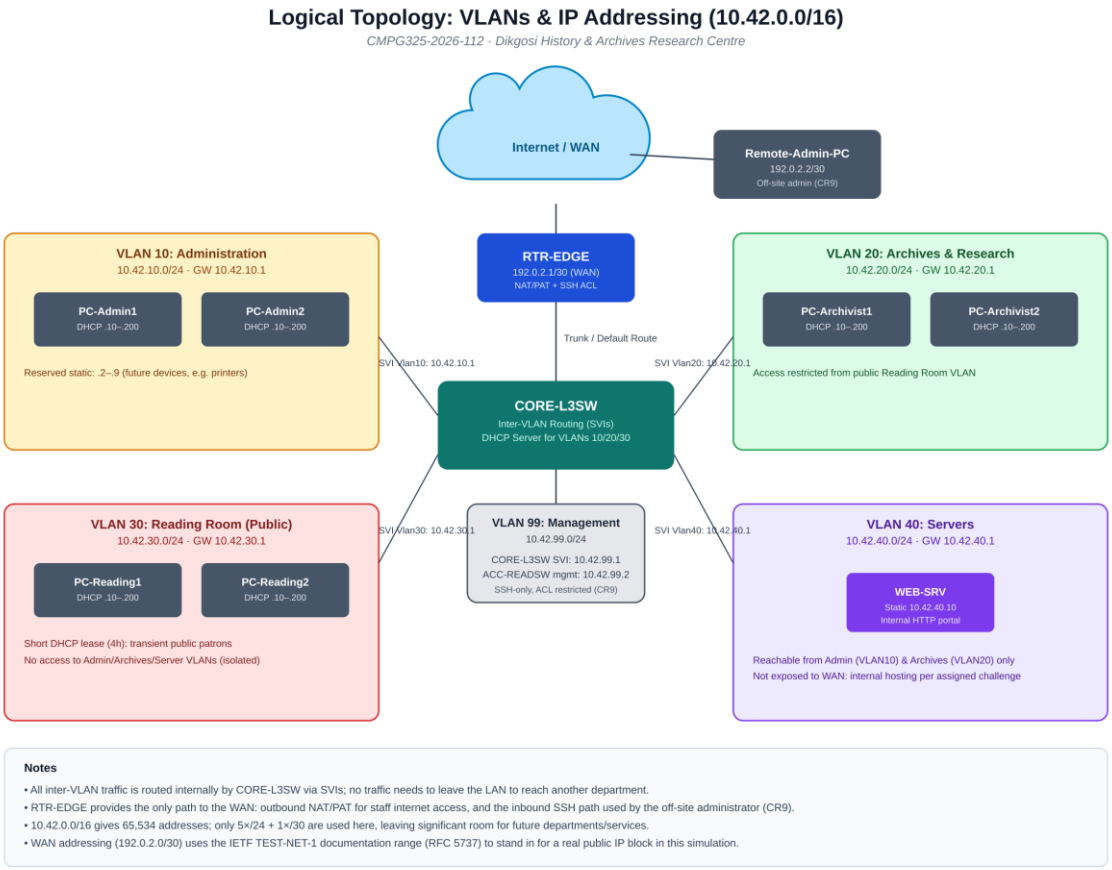


Figure 2. Logical topology: VLANs, subnets, and gateway addressing

3.1 VLAN Segmentation

VLAN	Name	Rationale
10	Administration	Separate broadcast domain for admin/finance traffic
20	Archives & Research	Staff who need access to the internal web portal
30	Reading Room (Public)	Isolated: public/transient users, no internal access
40	Servers	Hosts the internal HTTP server; reachable only from VLANs 10/20
99	Management	Switch/router management interfaces, SSH-only

3.2 Assigned Networking Challenge: Internal HTTP/Web Server

- WEB-SRV sits in the dedicated Servers VLAN (40) with a static IP (10.42.40.10) and runs the Packet Tracer HTTP service, hosting an internal catalogue/portal page.
- Access is restricted to VLAN 10 (Administration) and VLAN 20 (Archives & Research). The Reading Room (VLAN 30) and the WAN cannot reach it, since the brief specifies internal web service hosting, not a public-facing site.
- This will be enforced at Milestone 2 with an extended ACL on CORE-L3SW's Vlan40 SVI, verified with ping/browser tests from each VLAN plus a documented negative test from the Reading Room.

4. IP Addressing Plan

The assigned block 10.42.0.0/16 (65,534 usable host addresses) is subnetted into five /24 networks (one per VLAN) plus a /30 for the WAN point-to-point link. This uses only a small fraction of the assigned /16, leaving room for future departments or services without re-addressing the existing network.

4.1 Subnet Table

VLAN	Name	Network	Usable Range	Gateway (SVI)
10	Administration	10.42.10.0/24	10.42.10.1 – .254	10.42.10.1
20	Archives & Research	10.42.20.0/24	10.42.20.1 – .254	10.42.20.1
30	Reading Room (Public)	10.42.30.0/24	10.42.30.1 – .254	10.42.30.1
40	Servers	10.42.40.0/24	10.42.40.1 – .254	10.42.40.1
99	Management	10.42.99.0/24	10.42.99.1 – .254	10.42.99.1
N/A	WAN (RTR-EDGE ↔ ISP)	192.0.2.0/30	192.0.2.1 – .2	n/a (point-to-point)

WAN addressing uses 192.0.2.0/24 (TEST-NET-1, RFC 5737) as a documentation stand-in for a real ISP-assigned public block, since 10.42.0.0/16 is a private range and cannot represent "the internet" side of the simulation.

4.2 Static / Reserved Assignments

Device	Interface	VLAN	IP Address
RTR-EDGE	Gi0/0 (WAN)	N/A	192.0.2.1/30
RTR-EDGE	Gi0/1 (trunk to CORE-L3SW)	N/A	no IP (trunk)
CORE-L3SW	Vlan10 (SVI)	10	10.42.10.1/24
CORE-L3SW	Vlan20 (SVI)	20	10.42.20.1/24
CORE-L3SW	Vlan30 (SVI)	30	10.42.30.1/24
CORE-L3SW	Vlan40 (SVI)	40	10.42.40.1/24
CORE-L3SW	Vlan99 (SVI, mgmt)	99	10.42.99.1/24
ACC-READSW	Vlan99 (mgmt)	99	10.42.99.2/24
WEB-SRV	NIC	40	10.42.40.10/24
Remote-Admin-PC	NIC (simulated public)	N/A	192.0.2.2/30

Addresses .2–.9 in each end-user VLAN (10, 20, 30) are reserved for future static devices (e.g. network printers) and excluded from the DHCP pool.

4.3 DHCP Pools

Pool Name	VLAN	Range	Gateway	Lease Time
ADMIN-POOL	10	10.42.10.10 – .200	10.42.10.1	24 h
ARCHIVE-POOL	20	10.42.20.10 – .200	10.42.20.1	24 h
READING-POOL	30	10.42.30.10 – .200	10.42.30.1	4 h

Reading Room leases are set shorter (4 hours) than staff leases, reflecting the transient nature of public patron sessions.

4.4 VLSM Justification

- Every department's realistic headcount fits comfortably within a /24 (254 hosts), so variable-length subnetting would not meaningfully save usable address space here.
- Uniform /24s are simpler to explain and verify on the video demonstration, reducing the risk of addressing errors under project time pressure.
- The /16 block is large enough that address-space efficiency is not a real constraint for this project; scalability is prioritised over conservation.
- The WAN link correctly uses a /30 (only 2 usable hosts needed), demonstrating VLSM understanding where it is appropriate.

Github repository link : https://github.com/TebelloQekisi/cmpg325-2026-112-dharc-network.git